

TEACHER KEY & DIAGNOSTIC GUIDE

Incoming DP1 • use the section profile, not merely the total

Purpose: locate where a student's chain breaks: phenomenon → system → entities → quantities → relationship → model → equation. Award alternatives when the physics and the boundary choice are explicit.

1 | DESCRIPTION / PSEUDO-EXPLANATION / EXPLANATION

6 MARKS

ITEM	KEY	WHY
1	D	A measured change with values and elapsed time.
2	P	"Cooling made it cool" merely repeats the outcome.
3	E	Names a system boundary, direction of transfer and temperature difference.
4	D	Reports a controlled change and observed response.
5	P	"Attracted" labels the motion without adding a testable mechanism or model.
6	E	Invokes a field model, an interaction and a dependence on separation.

Interpretation: 0-2 suggests causal language is being accepted by its shape; 3-4 shows an emerging distinction; 5-6 shows secure recognition. Do not require students to treat a model as certainty: an explanation remains conditional on assumptions and evidence.

2 | BUILD THE MODEL BEFORE THE EQUATION

8 MARKS

PROMPT	ACCEPTABLE ANSWER / MARKING
System boundary [1]	Any coherent choice. Typical: trolley is the system and Earth is outside, interacting gravitationally. Also accept trolley + Earth if the resulting boundary language is consistent.
Two direct quantities [2]	Distance and time. Accept ramp angle if the student explicitly treats it as measured. Speed is derived from distance and time in the data as written; force is not directly measured here.
Relationship [1]	Examples: distance travelled per equal interval increases with time; a steeper ramp produces a larger change in motion. No symbols required.
Control [1]	Same trolley, surface, release point/method, ramp length or timing method. One valid control earns the mark.
Clause split [2]	Description: <i>"The distance travelled in equal time intervals increased"</i> . Explanation: <i>"because the component of the gravitational interaction ... produced a resultant force."</i>
Challenge [1]	Any possible result incompatible with the stated relationship, for example no systematic change when angle increases, or a change that follows surface rather than angle under controlled conditions.

Strong diagnostic clue: students who list objects but cannot name a relation are still at the noun stage. Students who write a formula without choosing a system have skipped the modelling decision.

TEACHER KEY & DIAGNOSTIC GUIDE

Incoming DP1 • use the section profile, not merely the total

3 | READ THE RELATIONSHIP INSIDE THE UNITS

8 MARKS

IDEA	WORD RELATIONSHIP	UNIT / CONVERSION
spring stiffness	force per unit extension	N m^{-1}
pressure	force per unit area	$\text{N m}^{-2} = \text{Pa}$
mechanical energy transfer	force acting through / multiplied by distance	$\text{N m} = \text{J}$
prefixes	Convert the value and the unit together.	$200 \text{ g} = 0.200 \text{ kg}$ $30 \text{ mm} = 0.030 \text{ m}$

Source-deck corrections to preserve: a pascal is N m^{-2} , not N m^{-3} . Dimensional consistency is necessary, but it neither establishes the physical model nor supplies dimensionless constants such as 2π .

4 | LANGUAGE FORENSICS

6 MARKS

ITEM	FAMILY	MODEL REPAIR
A	substance	At steady state, charge is not used up in the lamp; the circuit transfers energy and the current describes charge flow rate.
B	agent	The fluid exerts forces on the cube's surfaces. Greater pressure below than above can produce an upward resultant.
C	attribute	A force is an interaction involving the trolley and another object or field; it is not cargo carried by the trolley.
D	process-state	Acceleration describes change of velocity. An object can accelerate while moving slowly, or accelerate opposite to its velocity and slow down.
E	model-object	The graph line represents a relationship between plotted quantities; it is not the trolley's spatial path unless those axes are spatial coordinates.

Marking: one mark for each correct family; one mark for any repair that removes the named confusion and states a physically defensible relation.

5 | EXIT WARRANT

2 MARKS

[1] A unit check can show that a proposed relationship is dimensionally consistent, or reject one that is inconsistent. **[1]** Precise rewrite: "The units are consistent, so the equation survives one necessary check; evidence, assumptions and model validity are still required."

PROFILE | USE THE PATTERN, NOT JUST 30

30 MARKS

DOMAIN	MARKS	IF $\leq$ HALF
Description / explanation	/6	Require paired sentences in practical work: first what changed; then the model claim and evidence.
System / quantities / relation	/8	Use the routine: boundary $\rightarrow$ entities $\rightarrow$ measured quantities $\rightarrow$ verbal relationship.
Unit grammar	/8	Read 'per', 'square' and prefixes aloud before substitution; convert to base SI explicitly.
Linguistic ontology	/6	Repair one sentence from each family before introducing the corresponding equation.
Epistemic warrant	/2	Separate consistency checks from experimental support and explanatory adequacy.